

Chapter 6.

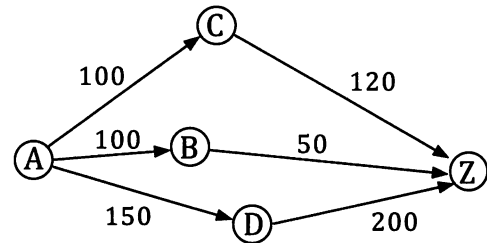
Maximum flow.

Situation

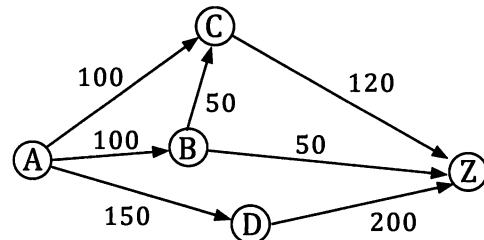
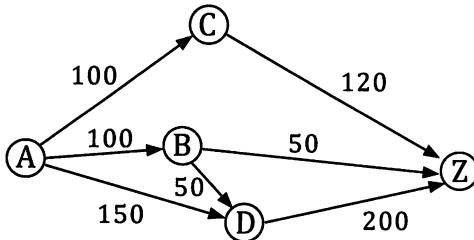
Suppose the network on the right shows the arrangement of water pipes around a farm with the numbers indicating the maximum rate of flow that each pipe can manage, in litres per minute.

Point A is the source of the water supply. The three pipes coming from A could draw water from A at a total rate of 350 litres every minute but could this amount of water be delivered to Z each minute?

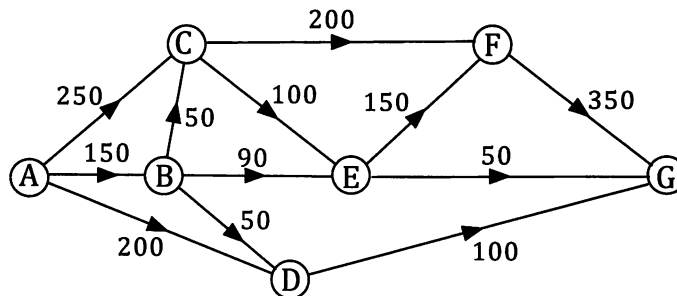
What is the maximum amount of water that can be delivered to Z each minute?



Now suppose we add a new pipe with a capacity of 50 litres per minute. The networks below show two ways that this could be done. What is the maximum flow from A, the **source**, to Z, the **sink**, in each of these situations?



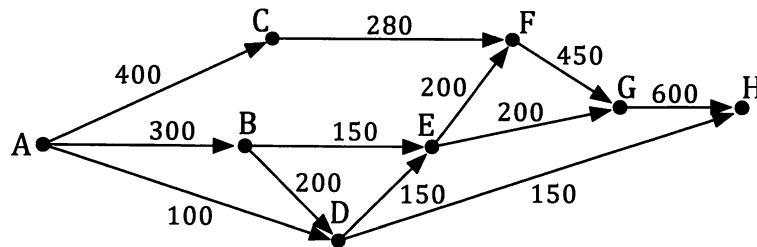
When the network is reasonably simple we can determine the maximum flow by mental reasoning as we have done above. However, for more complicated networks we need a procedure, or algorithm, to follow. Before such an algorithm is suggested for you, try to develop a procedure for yourself to determine the maximum flow through the network below from a source at A to a sink at G. The network shows the units of electrical power that can be delivered along the various links between relay stations.



Note: In reality, in situations like these there is often more than one *source* from which we can obtain the water or the power or whatever it is that is involved. In addition there is frequently more than one *sink* where this water or power is required. However, in this book we will restrict our attention to situations in which there is just one source and just one sink.

Maximum Flow – a systematic approach.

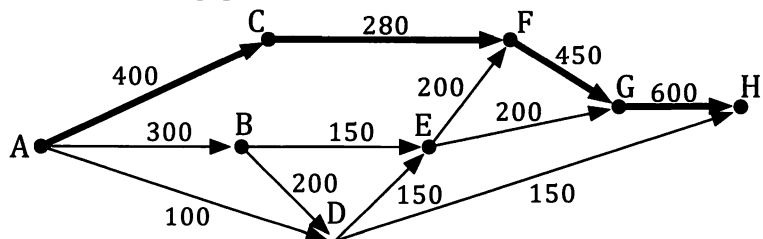
Suppose we wish to determine the maximum flow that can be achieved for the network shown below from the source at A to the sink at H.



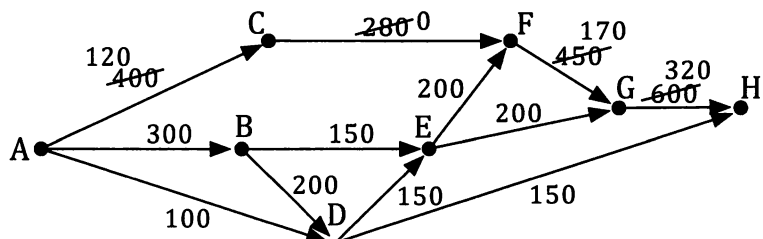
To approach the problem systematically we consider the separate routes in turn, from the top of the diagram to the bottom. We send the maximum flow we can along each route and note the remaining capacity of each pipe involved.

Consider route ACFGH.

This route can carry a maximum of 280 units (because that is all that CF can carry).



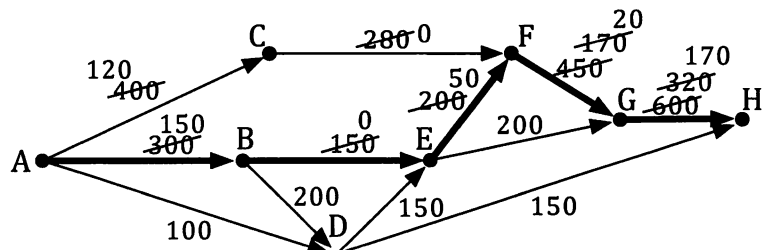
Sending 280 units along ACFGH leaves each pipe involved with the carrying capacity indicated in the next diagram. (Check carefully that you agree with these figures and can see how they are obtained.)



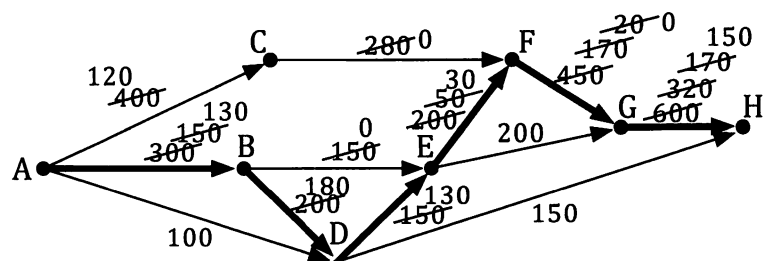
Now consider ABEFGH.

We can send 150 units along this route.

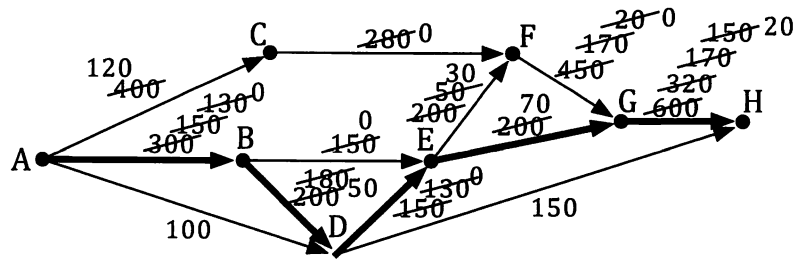
The remaining capacity of each pipe will then be as shown on the right



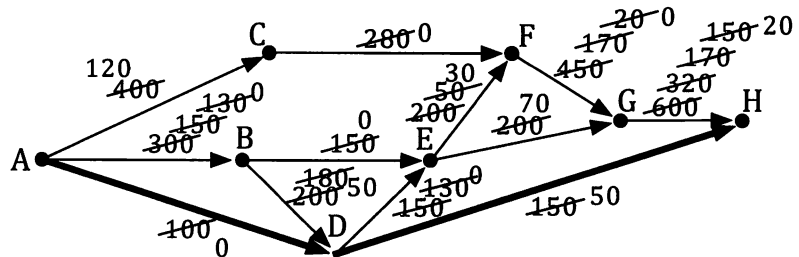
We can send 20 units along ABDEFGH.



We can send 130 units along ABDEGH.



We can send 100 units along ADH.



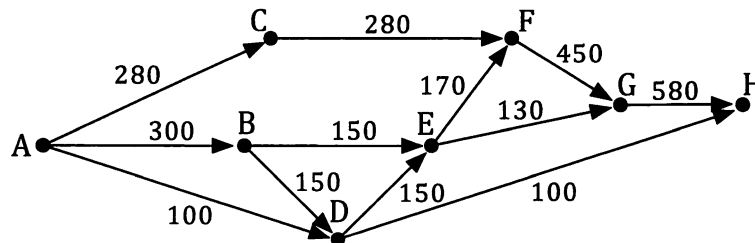
Thus the flow is:

ACFGH	280 units	+
ABEFGH	150 units	
ABDEFGH	20 units	
ABDEGH	130 units	
ADH	<u>100 units</u>	
	680 units	

giving a total of

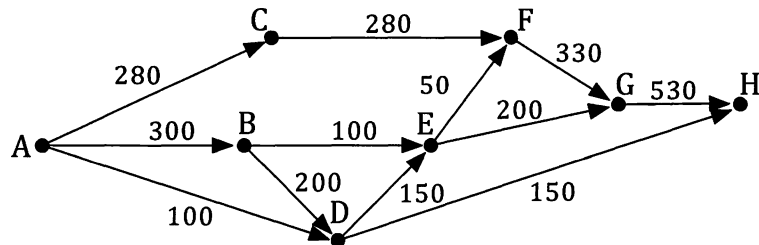
The maximum flow from the source at A to the sink at H is 680 units.

- Note:
- This process would normally be carried out using a single diagram as shown in the example on the next page.
 - The arcs leaving A have a total capacity of 800 units ($= 400 + 300 + 100$). Our maximum flow of 680 would indicate that there must be 120 units that we have not been able to use. This can be seen in the diagram above as 120 units of unused capacity in AC. Similarly the arcs going to the sink, H, have the capacity to deliver 750 units ($= 600 + 150$). Our maximum flow would indicate there must be 70 units that we have not been able to use. This can be seen in the diagram above as 50 units of unused capacity in DH and 20 units of unused capacity in GH.
 - By noting the unused capacity in each arc we can determine the flow along each pipe to obtain maximum flow:



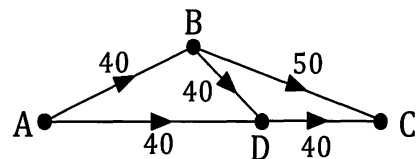
Notice in this diagram that 680 units (the maximum flow) leaves A, 680 units arrives at H and at all of the other vertices the total number of units entering is the same as the total number leaving.

- This diagram showing how the maximum flow can be achieved is not necessarily unique. If we use this method of considering the separate routes but start from the bottom route, ADH, we can arrive at the following arrangement for delivering the maximum flow. Note that the maximum flow is still 680 units but the way it is achieved is different.



- Considering separate routes, one after another, will not necessarily guarantee that we obtain the maximum flow.

For example in the network on the right, if we were to consider route ABDC first and direct 40 units along this route, we obtain a total flow of just 40 units, which is clearly not the maximum. To direct 40 units along ABDC

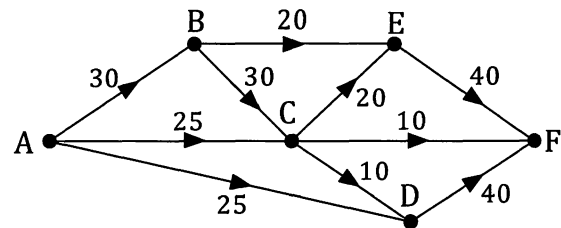


in this way is clearly not the wisest thing to do but in some more complicated networks the "wisest" strategies may not be so obvious. However, the method outlined here, with a little common sense applied, should give the maximum flow for most networks you are likely to meet in this course. Later in this chapter, we will see a way of checking our value to see if it is the maximum flow.

Example 1

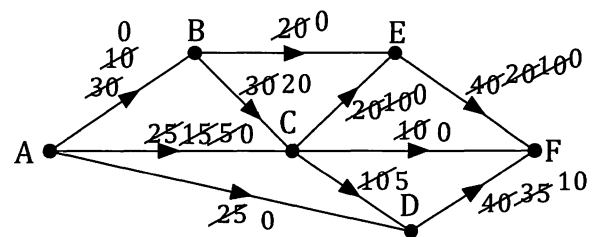
Determine the maximum flow possible for the network of pipes shown on the right with point A as the source and point F the sink.

Re-draw the network showing the flow you would direct along each pipe in order to achieve the maximum flow at F.

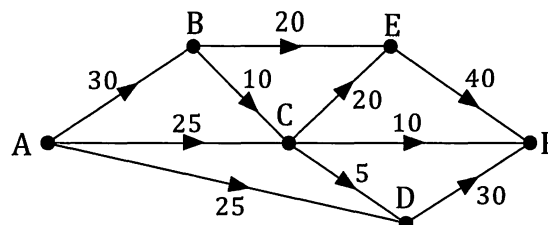


Using the systematic approach:

ABEF: 20 units
 ABCEF: 10 units
 ACEF: 10 units
 ACF: 10 units
 ACDF: 5 units
 ADF: 25 units Total: 80 units



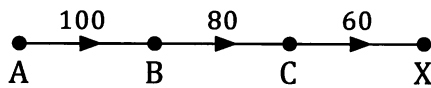
The maximum flow from A to F is 80 units and can be achieved as follows:



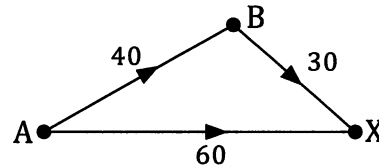
Exercise 6A

For each of the networks shown in questions 1 to 4 write down the maximum flow from the source at A to the sink at X (any working should be done mentally).

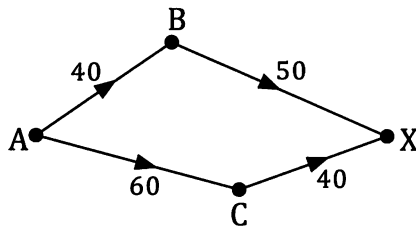
1.



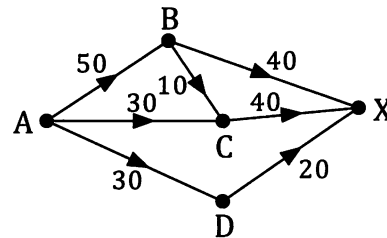
2.



3.

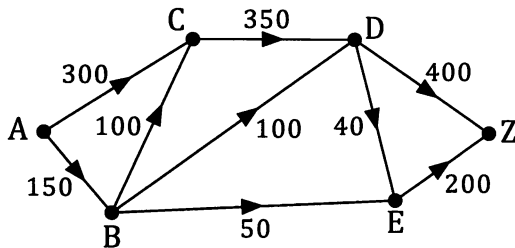


4.

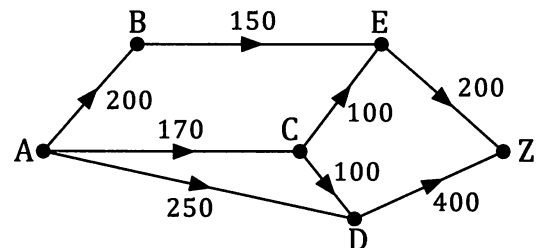


For each of the networks shown in questions 5 to 10 use a systematic approach to determine the maximum flow from the source at A to the sink at Z.

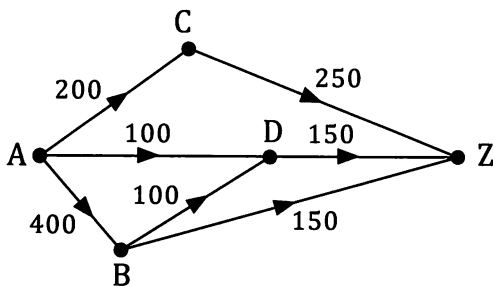
5.



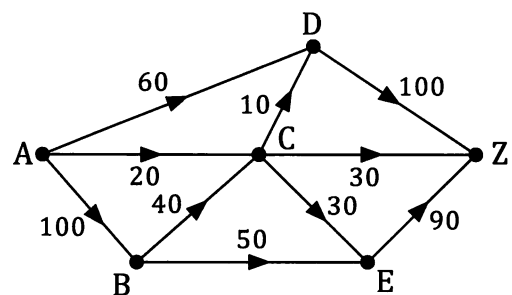
6.



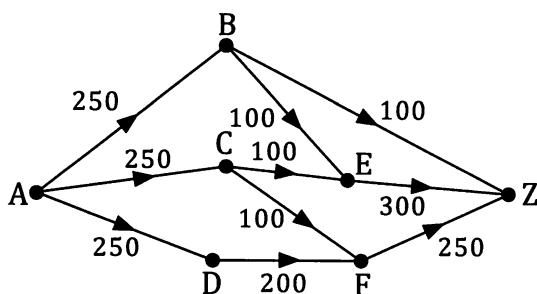
7.



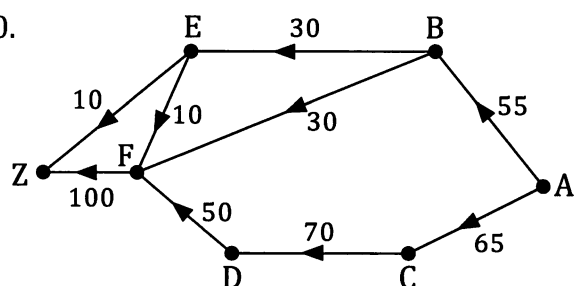
8.



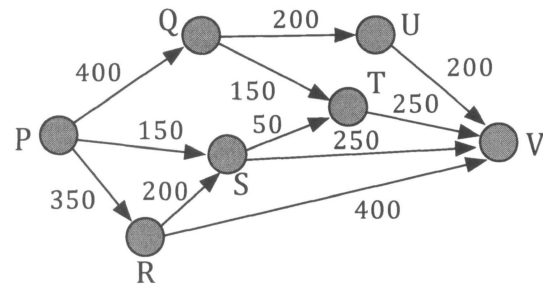
9.



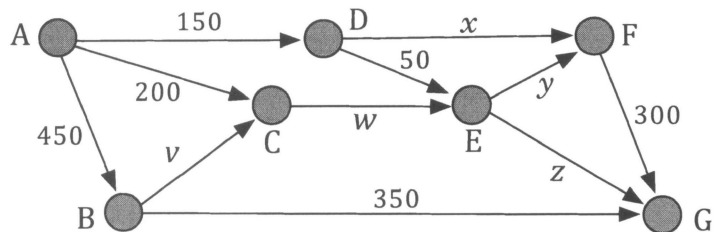
10.



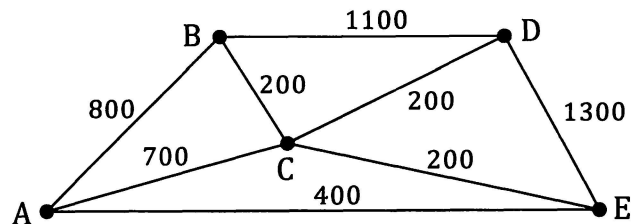
11. For the network shown on the right determine the maximum flow possible from the source at P to the sink at V. Re-draw the network showing the flow you would direct along each arc in order to achieve the maximum flow at V.



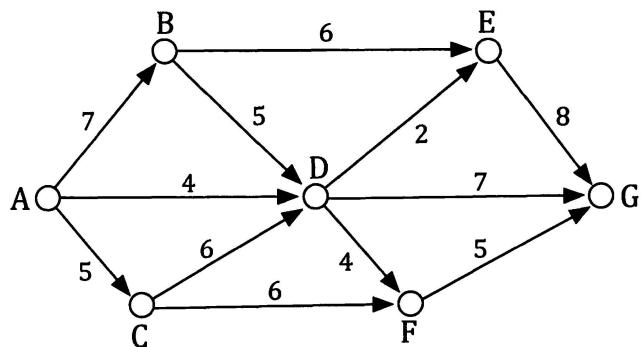
12. Given that the network shown on the right shows the flow along each arc that would achieve the maximum flow from the source at A to the sink at G, determine the values of v , w , x , y and z .



13. The network shown on the right represents a telecommunications system linking five locations, A to E. The number on each edge represents the maximum number of connections that can be active along that edge simultaneously. What is the maximum number of simultaneous connections that can exist from location A to location E? Draw a diagram showing the number of connections you would make along each edge, and show the directions of those connections, in order to produce this maximum situation.



14. The network shown on the right represents a system of water pipes connected to a main supply at A. The arcs in the network represent the pipes in the system and the numbers on the arcs give the maximum capacity of that section of pipe in units per minute.



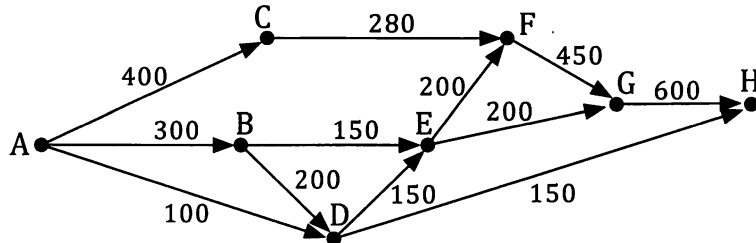
What is the maximum amount of water that could be delivered at point G per minute given that the only restriction is the capacity of the pipes?

One of the three 5 units per minute pipes is to be upgraded to an 8 units per minute pipe. What would be the effect of each of these three possible upgrades on the maximum flow at G?

Maximum flow = Minimum cut.

Earlier it was shown that by choosing an inappropriate route to start with, the total flow we end up with may not be the maximum flow possible. If the network is very complicated mistakes can easily be made. With careful use of our systematic method it will give us the maximum flow for the networks we are likely to meet on this course but we can check that it is the maximum flow by considering "cuts".

Consider the network shown on the right.



First draw vertical lines through the source on the left and through the sink on the right. A **cut** is then made by starting above the network and drawing a line that cuts only arcs of the network and finishes below the network. (The cut disconnects source from sink.)

The value of the cut is found by summing the cut arcs.

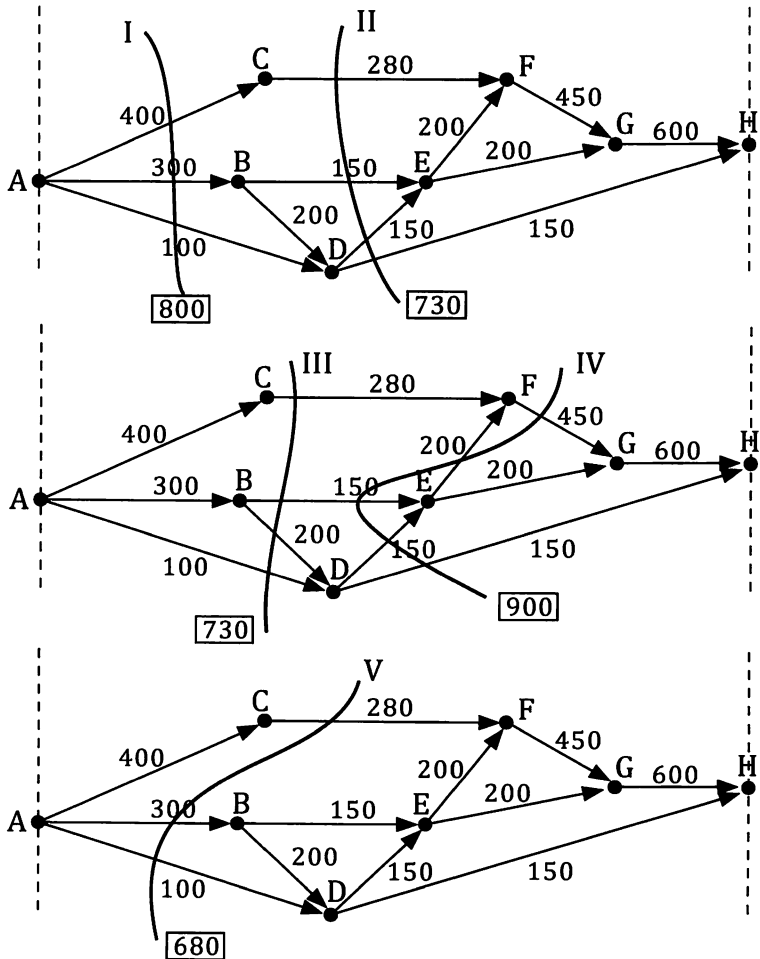
The diagram on the right shows two such cuts, I and II, and their values.

The next diagram shows two more cuts, III and IV.

Note that IV cuts EF from "below to above" (imagine EF horizontal with its direction of flow being from left to right). This is not included when determining the value of the cut. (The value comes from those arcs that are cut from "above to below".)

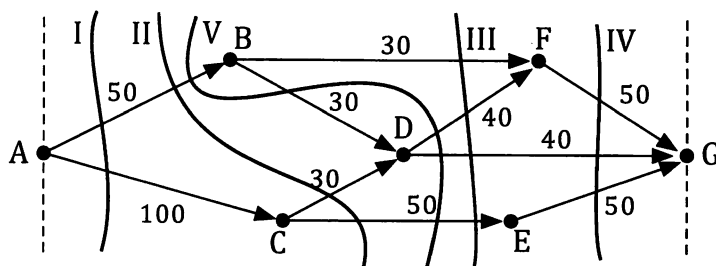
We can use the fact (not proved here) that **the maximum flow equals the minimum cut** to check our maximum flow. A cut value and a flow value can only be the same when the cut value is a "min" and the flow value a "max". Hence, if we can find a cut equal to our maximum flow then our value must indeed be the maximum.

When we met this network earlier we calculated the maximum flow as 680 units. We can find a cut having this value (see diagram) thus confirming our maximum flow value.

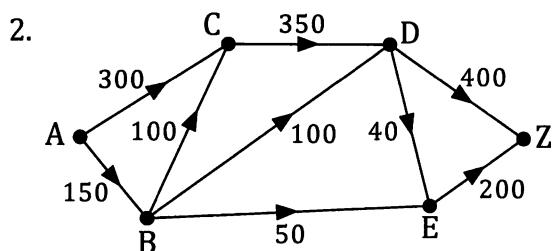


Exercise 6B

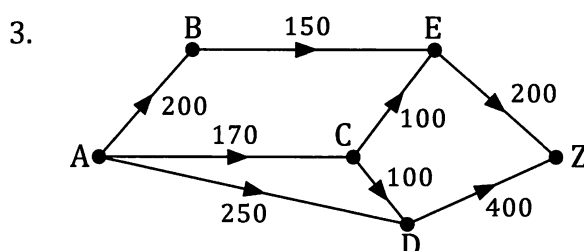
- Determine the value of the cuts I to V shown in the diagram on the right.



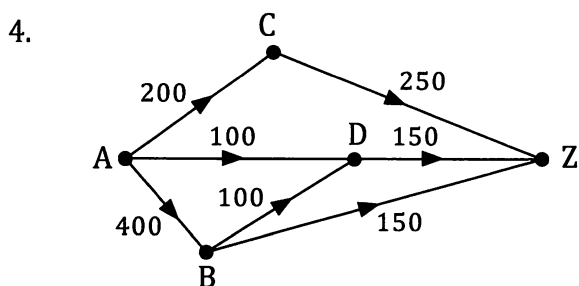
The networks below were all encountered in Exercise 6A. For each network confirm the stated maximum flow by finding a cut of the same value.



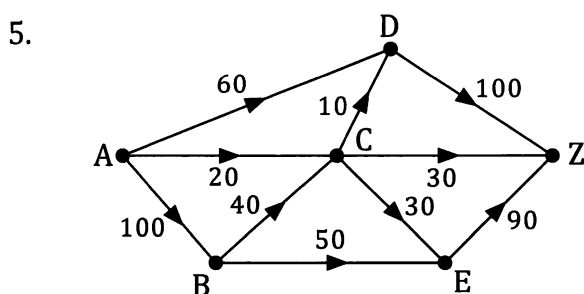
Maximum flow 450 units



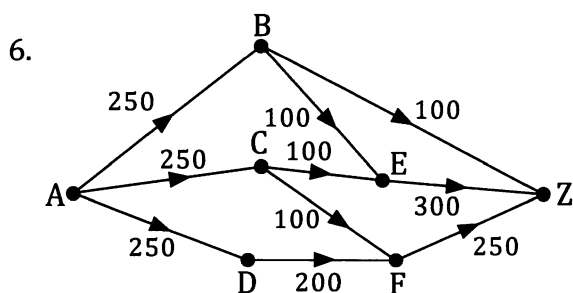
Maximum flow 550 units



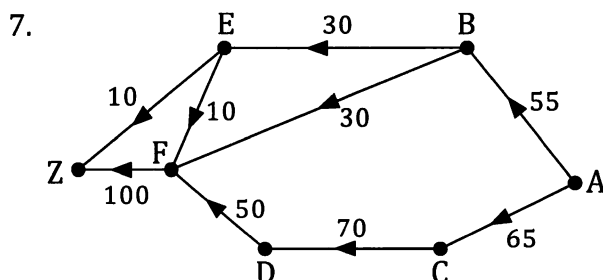
Maximum flow 500 units



Maximum flow 170 units



Maximum flow 550 units



Maximum flow 100 units

Miscellaneous Exercise Six.

This miscellaneous exercise may include questions involving the work of this chapter, the work of any previous chapters, and the ideas mentioned in the preliminary work section at the beginning of the book.

1. Apply the average (mean) percentage method explained earlier in this book to the data given below to determine the seasonal index for each quarter, Q1, Q2, Q3 and Q4, and explain what it is that these indices tell us. Give the indices as percentages rounded to the nearest whole percent.

Year One				Year 2			
Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
688	498	248	406	733	632	352	483

2. Find the length of the minimum spanning tree for the network defined by the table on the right. (A "-" in the table indicates there is no direct route between the locations.)

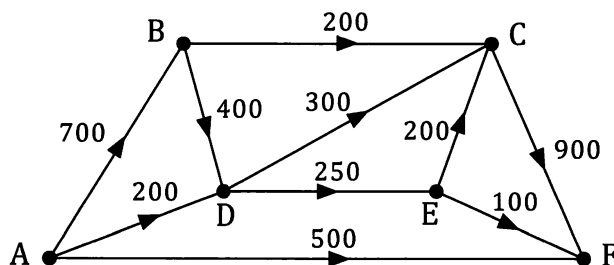
	A	B	C	D	E	F	G	H
A	-	37	70	68	-	80	-	-
B	37	-	40	-	-	-	58	-
C	70	40	-	-	-	-	52	-
D	68	-	-	-	17	-	36	-
E	-	-	-	17	-	20	-	67
F	80	-	-	-	20	-	-	78
G	-	58	52	36	-	-	-	39
H	-	-	-	-	67	78	39	-

3. (Use a calculator with a built in facility to calculate compound interest calculations or a suitable online finance calculator.)
A loan of \$500 000 is taken out to finance the purchase of a house. Interest is charged at 8.5% per annum with interest compounded monthly.
- (a) How much must be repaid per month if the loan is to be paid off in 25 years?
(b) What is the total amount of interest paid in the 25 years? (Nearest \$10.)
4. A "one off" investment to open a savings account, in which interest is calculated annually, is such that if n years later the value of the account is $\$T_n$ then the value one year after that, $\$T_{n+1}$, is such that

$$T_{n+1} = T_n \times 1.072 \text{ and } \$T_0 = \$12000.$$

- (a) What is the initial investment?
(b) What is the annual interest rate?
(c) Using a recursive approach and not the financial capability of some calculators and computer programs, determine the value of the account after six years.
5. Under a reducing balance (fixed percentage) depreciation scheme of 12% per annum how much does an asset initially worth \$120 000 depreciate in each of its first three years?

6. The diagram below shows the pipe system connecting a number of watering points on a farm to the main source, A. The number on each section indicates the maximum flow that pipe can manage in litres per minute.



What is the maximum number of litres per minute that could be delivered at outlet F under this system? Make a copy of the diagram and indicate the flow along each pipe (in L/min) to achieve this maximum delivery at F.

7. The following table shows the number of short term visitor arrivals into Australia for each four month period from Jan-Feb-Mar-Apr 2008 to Sept-Oct-Nov-Dec 2013.

n	Months	Year	Number of short term visitors	3 pt MA (M)
1	Jan-Feb-Mar-Apr	2008	1898800	—
2	May-Jun-Jul-Aug	2008	1691100	
3	Sept-Oct-Nov-Dec	2008	1922500	
4	Jan-Feb-Mar-Apr	2009	1872200	
5	May-Jun-Jul-Aug	2009	1587200	
6	Sept-Oct-Nov-Dec	2009	2030800	
7	Jan-Feb-Mar-Apr	2010	1933600	
8	May-Jun-Jul-Aug	2010	1720400	
9	Sept-Oct-Nov-Dec	2010	2136100	
10	Jan-Feb-Mar-Apr	2011	1959800	
11	May-Jun-Jul-Aug	2011	1717700	
12	Sept-Oct-Nov-Dec	2011	2093300	
13	Jan-Feb-Mar-Apr	2012	2023100	
14	May-Jun-Jul-Aug	2012	1777600	
15	Sept-Oct-Nov-Dec	2012	2231500	
16	Jan-Feb-Mar-Apr	2013	2103100	
17	May-Jun-Jul-Aug	2013	1895700	
18	Sept-Oct-Nov-Dec	2013	2381700	—

[Source of data: Australian Bureau of Statistics.]

With the assistance of a computer spreadsheet, or otherwise, :

- Determine the entries for the final column, the 3 point moving averages, giving values to the nearest 1 hundred.
- Produce a graph with n on the horizontal axis and displaying both the raw data and the moving average data as time series line graphs.
- Use linear regression to obtain the equation of the line of best fit for predicting M , the moving average, given n , i.e. the equation $M = an + b$. (Give a to the nearest 10 and b to the nearest 1000.)
- Interpret the value of " a " in the equation from part (c).